

Venkata Charan Bommineni

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Data Engineer



Highly skilled Data Engineer with **5+ years** of experience in developing robust data architectures and implementing end-to-end data processing workflows. Excited to contribute my expertise in SQL, Python, and cloud Technologies.

PROFILE SUMMARY

- Around **5+** years of experience in **Data engineering** encompassing **Requirements Analysis, Design Specification, and Testing** in both **Waterfall** and **Agile** methodologies.
- Hands-on experience with **Amazon EC2, Amazon S3, Amazon RDS, VPC, IAM, Amazon Elastic Load Balancing, Auto Scaling, CloudWatch, SNS, SES, SQS, Lambda, EMR** other services of the **AWS** family.
- Worked with **csv, Avro, Parquet** data to load into **Dataframes** and do the analysis. Hands on experience on **Google Cloud Platform (GCP)** in all the big data products **BigQuery, Cloud Data Proc, Google Cloud Storage, Composer (Air Flow as a service)**.
- Proficiency with **Scala, Apache HBase, Hive, Pig, Sqoop, ZooKeeper, Spark, Spark SQL, Spark Streaming, Kinesis, Airflow, Yarn, and Hadoop (HDFS, MapReduce)**. Implemented **Big Data** solutions using **Hadoop** technology stack, including **Pyspark, Hive, Sqoop, Avro** and **Thrift**.
- Working knowledge of **HDFS, Kafka, Map Reduce, Spark, PIG, HIVE, Sqoop, HBase, Flume, and Apache ZooKeeper** as tools for designing and deploying end-to-end big data ecosystems.
- Hands-on experience with **Spark, Databricks, and Delta Lake** and Practical experience with **Python** and **Apache Airflow** to create, schedule, and monitor workflows. Experience in automating day-to-day activities by using **Windows PowerShell**.
- Experienced working in both **Agile** and **Waterfall** based development environment and participating in **Scrum** sessions.
- Practical experience with **Python** and **Apache Airflow** to create, schedule, and monitor workflows. Experienced in building **Snow Pipes**, migrating **Teradata** objects into **Snowflake** environment.
- Proficient with Container systems like **Docker** and Container orchestration like **EC2 Container Service, Kubernetes**, worked with **Terraform**.
- Experience in implementing Azure data solutions, provisioning storage account, **Azure Data Factory, SQL server, SQL Databases, SQL Data warehouse, Azure Data Bricks and Azure Cosmos DB**.
- Expert in designing Parallel jobs using various stages like **Join, Merge, Lookup, remove duplicates, Filter, Dataset, Lookup file set, Complex flat file, Modify, Aggregator, XML**.
- Hands-on experience using **Snowflake** utilities, **SnowSQL, SnowPipe, Big Data model techniques using Python / Java**.
- Expertise in building **CI/CD** on **AWS** environment using **AWS Code Commit, Code Build, Code Deploy and CodePipeline** and experience in using **AWS CloudFormation, API Gateway, and AWS Lambda** in automation and securing the infrastructure on **AWS**.
- Proficient in building **CI/CD** pipelines in **Jenkins** using pipeline syntax and **groovy** libraries and Excellent knowledge on **Hadoop** Architecture and ecosystems such as **HDFS, Hive, Pig, Sqoop, Job Tracker, Task Tracker, Name Node, Data**
- Involved in using **Jenkins, Selenium, Docker** and **Kubernetes** clusters and **power Shell scripting** for efficient data management and decentralized access.
- Experience working with **Azure Logic APP** Integration tool and working with Data warehouse like **Oracle, SAP, HANA**.
- Instantiated, created, and maintained **CI/CD** (continuous integration & deployment) pipelines and apply automation to environments and applications and worked on various automation tools like **GIT, Terraform, Ansible**.
- Developing serverless **AWS Lambda** functions for real-time data processing, automation of ETL workflows, and integration with other **AWS** services like **S3, DynamoDB, and Kinesis** to streamline data pipelines and enhance system scalability.

EDUCATION

Tennessee Wesleyan University, Masters in Business Analytics, USA from 2024

TECHNICAL SKILLS

Programming Languages:

- **Python:** Widely used for scripting, data manipulation, and developing data pipelines.
- **SQL:** Essential for querying and managing relational databases.
- **Java/Scala:** Useful for working with big data tools like Apache Spark and Hadoop.

Data Warehousing & Databases:

- **Relational Databases:** MySQL, PostgreSQL, SQL Server.
- **NoSQL Databases:** MongoDB, Cassandra, HBase.
- **Data Warehousing:** Amazon Redshift, Google BigQuery, Snowflake.

Big Data Technologies: HDFS, Hue, MapReduce, PIG, Hive, HCatalog, HBase, Sqoop, Impala, Zookeeper, Flume, Kafka, Yarn, Cloudera Manager, Kerberos, PySpark Airflow, Kafka, Snowflake Spark Components

Cloud Platforms:

- **AWS:** Services like S3 (storage), Redshift (data warehousing), Lambda (serverless computing), and EMR (Hadoop/Spark).
- **Azure:** Services like Azure Data Lake, Azure Synapse Analytics, and Azure Databricks.
- **Google Cloud Platform (GCP):** BigQuery, Cloud Dataflow, and Cloud Storage.

Visualization& ETL tools: Tableau, PowerBI, Informatica, Talend

Web/Application server: Apache Tomcat, WebLogic, WebSphere

Version controls and Tools: GIT, Maven, SBT, CBT

Data Processing Frameworks: Apache Hadoop, Apache Spark, Apache Flink, Apache Beam

WORK EXPERIENCE

CNA Financial Corporation, Chicago, Illinois, USA

Feb 2024 - Present

Azure Data Engineer

Description: CNA Financial Corporation is a prominent commercial insurance company specializes in a wide array of insurance products tailored for businesses and professionals. I am focusing on designing and implementing scalable and secure data processing pipelines using Azure Data Factory and other Azure services and configuring data connectors, and ensuring data integrity during the ingestion process and development of data processing pipelines

Responsibilities:

- Implemented **Azure Data Factory** (ADF) extensively for ingesting data from different source systems like relational and unstructured data to meet business functional requirements
- Worked with complex **SQL** views, Stored Procedures, Triggers, and packages in large databases from various servers. Used **Azure Devops & Jenkins** pipelines to build and deploy different resources (Code and Infrastructure) in Azure. Ensure the developed solutions are formally documented and signed off by business.
- Develop data pipelines using **Azure Data Factory, Azure Databricks, and Azure Synapse Analytics** for **ETL** processes.
- Develop and maintain robust **ETL** processes to extract, transform, and load data from various sources into **Azure Data Lake, Azure SQL Database, or Azure Synapse**.
- Optimize data storage solutions using **Azure Data Lake Storage, Azure SQL Data Warehouse, and Azure Cosmos DB**.
- Work closely with security teams to ensure data solutions adhere to banking industry standards and regulatory requirements (e.g., **GDPR, CCPA**).
- Utilize **Azure Monitor** and **Azure Log Analytics** for performance monitoring and troubleshooting.
- Experience working with **Docker** to improve our **(CI/CD) Continuous Delivery framework** to streamline releases
- Wrote **Spark**-Streaming applications to consume the data from **Kafka** topics and write the processed streams to **HBase**.
- Created interactive and insightful **Tableau** dashboards that visualized key performance metrics, enabling business stakeholders to make informed decisions.
- Used **Pig** as **ETL tool** to do Transformations with joins and pre-aggregations before storing the **data** onto **HDFS** and assisted Manager by providing automation strategies, **Selenium/Cucumber** Automation and **JIRA** reports.

- Configured and scaled **SaaS** applications on **Azure** to meet customer demand, using tools such as **Azure Auto scale** and **Azure Monitor**. Managed, Configured and scheduled resources across the cluster using **Azure Kubernetes Service**.
- Built **PySpark** code in validate the data from raw source to **Snowflake tables**. created and triggered automated builds and **Continuous Deployments (CI/CD)** using **Jenkins**.
- Monitored **Spark Cluster** using Log Analytics and **Ambari Web UI**. Transitioned log storage from **Cassandra** to **Azure SQL** Data warehouse and improved the query performance. Involved in developing data ingestion pipelines on **Azure HD Insight Spark Cluster** using **Azure Data Factory** and **Spark SQL**

Environment: Azure Data Factory, SQL, Azure Devops, Jenkins, Azure Databricks, Azure Synapse, ETL, GDPR, CCPA, Docker, CI/CD, Kafka, HBase, Selenium, Cucumber, JIRA, Azure Kubernetes Service, Snowflake, PySpark, Ambari Web UI, Cassandra, Spark SQL.

Medline Industries, Northfield, Illinois, USA

July 2023 - Jan 2024

AWS Data Engineer

Description: Medline Industries is a privately held healthcare company. Designed, constructed, installed, tested, and maintained data management systems and implementing security measures to protect data from abuse or illegal access and Troubleshooting data-related issues and proposing solutions for improvements.

Responsibilities:

- Design, develop, and maintain **data pipelines** and **ETL processes** using **AWS Glue**, **AWS Lambda**, and **Apache Spark** to handle large-scale healthcare data.
- Build and optimize **data warehouses** using **Amazon Redshift & Amazon S3** for storing and querying structured healthcare datasets.
- Develop and manage real-time data streaming processes using **Amazon Kinesis** and **Apache Kafka** to facilitate timely healthcare data analytics and collaborate with data scientists and analysts to create scalable solutions using **AWS Sage Maker** for predictive analytics and machine learning in healthcare.
- Automate deployment processes using **AWS Code Pipeline**, **AWS Code Build**, and **Terraform** to manage infrastructure as code and monitor and troubleshoot data workflows with **AWS CloudWatch**, ensuring high availability and reliability of healthcare applications.
- Perform data transformation and validation tasks using **Python**, **SQL**, and **Pyspark** to ensure data quality.
- Design secure API endpoints using **AWS API Gateway** and integrate with **AWS Lambda** for scalable healthcare application interactions and Leverage **AWS Athena** for ad-hoc querying of healthcare data stored in **Amazon S3**.
- Optimize and manage databases such as **Amazon Aurora**, **DynamoDB**, and **PostgreSQL** for efficient healthcare data storage.
- Facilitate migration of on-premises healthcare data to AWS using **AWS Snowflake**, **AWS Database Migration Service (DMS)**, and **AWS Data Sync** and compliance through encryption and monitoring solutions like **AWS Macie** and **AWS Shield**.
- Perform big data analytics on healthcare data using **AWS EMR** integrated with **Hadoop** and **Hive**.
- Collaborate with DevOps teams to establish **CI/CD pipelines** and containerize applications using **Docker** and **Kubernetes** on **Amazon ECS** or **EKS**.
- Develop and optimize **ETL workflows** using **AWS Glue**, **Apache Airflow**, **Talend**, and **Informatica** to extract, transform, and load large-scale healthcare data from diverse sources into centralized data warehouses.

Environment: AWS Glue, AWS Lambda, Apache Spark, Amazon Redshift, Amazon S3, Amazon Kinesis, AWS Sage Maker, Terraform, Python, SQL, Pyspark, AWS API Gateway, AWS Lambda, Amazon Aurora, DynamoDB, PostgreSQL, AWS Snowflake, Hadoop, Hive, CI/CD pipelines, Docker, Kubernetes, Amazon ECS, Apache Airflow, Talend, Informatica

Kotak Mahindra Bank, Hyderabad, India

Sep 2021 - July 2022

Role: GCP Data Engineer

Description: Kotak Mahindra Bank is one of India's leading private sector banks provides a comprehensive range of banking products and services. Involved in designing and modernizing data solutions on GCP by integrating third-party data technologies and native GCP services and Using cloud services and tools to automate data workflows.

Responsibilities:

- Design and implement **data pipelines** using **Google Cloud Dataflow**, **Apache Beam**, and **Cloud Composer** to process and transform large-scale banking data.
- Perform **ETL/ELT operations** using **SQL**, **Python**, and **Big Query ML** to prepare data for downstream analytics and machine learning models.
- Utilize **Google Data Studio** and **Looker** for creating interactive dashboards and reports for banking executives and stakeholders.
- Automate deployment and infrastructure management using **Terraform**, **Cloud Deployment Manager**, and **Jenkins**.
- Collaborate with DevOps teams to containerize applications using **Docker** and orchestrate them on **Google Kubernetes Engine (GKE)**.
- Leverage **Cloud Dataproc** for big data analytics and processing using **Hadoop**, **Hive**, and **Spark** in banking use cases.
- Develop and optimize **data warehouses** on **Big Query** for efficient querying and reporting of transactional and analytical banking data.
- Manage and monitor **data orchestration workflows** using **Apache Airflow** on **Cloud Composer**.
- Develop interactive dashboards and reports using **Power BI** by integrating data from **Snowflake**, enabling real-time insights into banking transactions, customer analytics, and financial performance.
- Design and optimize complex **SQL queries**, stored procedures, and views using **MySQL**, **PostgreSQL**, **SQL Server**, and **Oracle SQL** to support banking data analysis and reporting.

Environment: Google Cloud Dataflow, Apache Beam, Cloud Composer, SQL, Python, Big Query ML, Terraform, Cloud Deployment Manager, Jenkins, Google Data Studio, Looker, Docker, Google Kubernetes Engine, Hadoop, Hive, Spark, Apache Airflow, Power BI, Snowflake, PostgreSQL, Oracle.

Glenmark Pharmaceuticals, Hyderabad, India

June 2020 – Aug 2021

Data Engineer

Description: Glenmark Pharmaceuticals is a global pharmaceutical company focuses on developing and manufacturing a wide range of pharmaceutical products. Build and maintain data infrastructures to ensure data is accessible and compliant with regulations and create pipelines that convert raw data into usable formats for data scientists.

Responsibilities:

- Design and implement **ETL pipelines** using tools like **Apache Nifi**, **Talend**, or **Informatica** to process and transform clinical and pharmaceutical data.
- Utilize **Hadoop Distributed File System (HDFS)** for distributed storage of large datasets generated from research and clinical trials and Process large-scale data using **Apache Spark** and **Databricks** for real-time and batch analytics on pharmaceutical manufacturing data and develop and maintain NoSQL databases such as **Cassandra**, **HBase**, or **MongoDB** to handle unstructured data like genomic sequences or patient records. Design interactive dashboards using **Tableau**, **Power BI**, or **Looker** to visualize drug trial results, sales performance, and supply chain metrics.
- Monitor data pipeline performance using tools like **Prometheus**, **Grafana**, or **Google Cloud Monitoring** and Utilize **Git** for version control and **Maven** for build automation to streamline the development lifecycle of data engineering projects.
- Create and manage **CI/CD pipelines** using tools like **Jenkins**, **GitLab CI**, or **Azure DevOps** to automate deployments and store and manage research data in **Google Cloud Storage**, **Azure Blob Storage**, or **HDFS** for secure and cost-efficient storage solutions and develop and manage data ingestion pipelines with **Zookeeper** to coordinate distributed systems and maintain consistency and Implement data lineage and governance solutions using tools like **Collibra** or **Informatica** to ensure transparency and traceability.
- Develop and maintain automation scripts for data extraction, transformation, and loading (ETL) processes using **Python** and **Shell scripting** and Develop data monitoring dashboards using **Prometheus** and **Grafana** to ensure system health

ENVIRONMENT: Apache Nifi, Talend, Informatica, Google Big Query, Azure Synapse Analytics, Snowflake, Apache Spark, Cassandra, HBase, MongoDB, Tableau, Power BI, Looker, Apache Kafka, Prometheus, Grafana, Maven, CI/CD, Jenkins, Zookeeper, Collibra, Informatica.